



Alberta Public Health Disease Management Guidelines

Anthrax



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Revisions

Revision Date	Document Section	Description of Revision
November 2024	General	Entire guideline updated to align with current information.
	Case Definition	- Molecular testing moved from Probable to Confirmed Case. - Probable Case definition removed.
	Reporting Requirements	- Negative lab results also now reportable (i.e., all results) by laboratories. - CFIA and IHR reporting requirements added.
	Management of Contacts	Added PEP and vaccine info.

Case Definition

Confirmed Case

Clinical illness^(A) with laboratory confirmation of infection (pre- or post-mortem):

- Isolation of *Bacillus anthracis* from an appropriate clinical specimen (e.g., blood, sputum, pus, fluid or swab from lesion)^(B);

OR

- Detection of *B. anthracis* DNA by molecular testing (e.g., polymerase chain reaction [PCR] testing) in an appropriate clinical specimen^(B) (e.g., blood)

Suspect Case

Clinical illness^(A) in a person who is epidemiologically linked^(C) to a confirmed or suspected animal case or contaminated animal product.

^(A) Clinical illness: Four distinct clinical forms including cutaneous, inhalation, gastrointestinal/ingestion, and injection. See [Table 1: Clinical Presentation](#) for details on specific symptoms for each form.

^(B) Refer to the [Alberta Precision Laboratories Test Directory](#) and [National Microbiology Laboratory \(NML\) Guide to Services](#) for current specimen collection and submission information.

^(C) Epidemiological link includes an exposure to an environment, food, animal, or other material and objects which is suspected or confirmed to be contaminated with *B. anthracis*.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Zone Medical Officer of Health (MOH) (or designate), of all confirmed and suspect cases in the prescribed form by the Fastest Means Possible (FMP).

Laboratories

- All laboratories, shall report all positive anthrax laboratory results, and negative results^(D) for suspect cases, by FMP to the:
 - Chief Medical Officer of health (CMOH) (or designate), and
 - Zone MOH (or designate).

Alberta Health Services and Indigenous Services Canada (ISC)

- The Zone MOH (or designate) of the zone where the case currently resides shall notify the CMOH (or designate) by FMP of all confirmed and suspect cases.
- The Zone MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and suspect cases to the CMOH (or designate) within one week of notification and the final NDR (amendment) within two weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by FMP:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Additional Reporting Requirements

Anthrax in animals is reportable to the Canadian Food Inspection Agency (CFIA), in accordance with the *Reportable Diseases Regulation* under the *Health of Animals Act (Canada)*. Animal health issues associated with the source of human anthrax disease should be reported to the District Veterinarian of the CFIA by the Zone MOH and to the Office of Chief Provincial Veterinarian by the CMOH. [CFIA contact information is available here.](#)

NOTE: The MOH does not need to report to CFIA if it is known that a private veterinarian is already involved and has reported the animal health concern to CFIA.

International Health Regulation Reporting

Under the [International Health Regulations \(IHR\) \(2005\)](#), enhanced reporting of anthrax to the World Health Organization (WHO) is mandatory. Alberta Health notifies the Public Health Agency of Canada (PHAC) of every confirmed or suspect case of anthrax and PHAC notifies the WHO of all confirmed human cases within 24 hours of initial notification.

^(D) Given the urgency/sensitivity of potential anthrax cases, negative results are also reportable by FMP.

Clinical Assessment and Epidemiology

Etiology

Bacillus anthracis is a gram-positive, encapsulated, spore-forming, non-motile rod bacteria.^(1,2)

Clinical Presentation

Anthrax is an acute bacterial zoonotic disease that can cause severe illness. Illness may present in one of four forms (Table 1), depending on the route of exposure.^(1,3) Fever and non-specific symptoms are common; systemic illness may include shock or meningitis.

Table 1: Clinical Presentations of Anthrax

ANTHRAX FORM	PRESENTATION
Cutaneous Anthrax	95% of all human cases.^(1,2) - Characterized by an itchy, pruritic papule, at the site of inoculation, progressing over 2 to 6 days to a painless, ulcerated lesion with considerable swelling, then into black eschar.^(1,3) - Most common sites of infection are the head, neck, forearms and hands.^(1,3) - Case fatality of untreated cases is 5–20%, but with effective treatment is <2%.^(1–3)
Gastrointestinal, Ingestion Anthrax	- Rare and difficult to identify, more commonly presenting as outbreaks associated with consumption of contaminated meat.^(1,3) - Symptoms include sore throat, difficulty swallowing, abdominal pain, fever, fatigue, headache, nausea, vomiting, and diarrhea which may contain blood.^(1–3) - The estimated case-fatality of gastrointestinal anthrax is >50%. With treatment approximately 60% survive.^(1,3)
Injection Anthrax	- Similar to cutaneous anthrax but occurs deep under the skin or muscle near the site of injection in drug users.⁽⁴⁾ - Symptoms include fever, blisters/sores at site of injection, with surrounding edema, progressing to a painless sore with black centre.^(1,2) - Case fatality rate is estimated at 20%.⁽¹⁾
Inhalation Anthrax	- The most severe, rarest and lethal disease form.^(2,4) - Characterized by fever, malaise, fatigue, cough, chest pain progressing to acute onset of respiratory distress and shock.^(1,4) - Case fatality is estimated to be > 85%.⁽¹⁾ Approximately 55% of patients survive with aggressive treatment.⁽³⁾

Diagnosis

Diagnosis is based on the demonstration of *B. anthracis* bacilli in culture or through molecular testing in samples from sites of infection: blood, sputum, or wounds.^(1,2) Samples should be taken prior to initiation of antimicrobial treatment, which reduces sensitivity of cultures.^(1,2) Chest X-rays or CT can evaluate for mediastinal widening or pleural effusion, common findings in cases of inhalation anthrax.^(1,2,5)

Treatment

- Anthrax requires urgent medical evaluation and treatment.^(1,4,6) Consultation with an infectious disease specialist is recommended.
- In cases of systemic anthrax infection, immediate access to critical care support and initiation of combination intravenous (IV) antimicrobial therapy are essential. Adjunctive antitoxin therapy may also be beneficial.^(1,2,7)
- Cutaneous lesions become non-infectious after 24 hours of appropriate treatment.⁽²⁾

Reservoir

Anthrax primarily affects domestic and wild herbivores (e.g., cattle, sheep, goats, camels, antelope, etc.) that ingest contaminated vegetation or water.^(1,2,8) Infected animals shed anthrax during death and decomposition, releasing spores into the environment. When exposed to air or nutrient deprivation, vegetative *B. anthracis* cells form spores, which contaminate the soil and allow the bacteria to persist.^(1,4,8) These dormant spores can remain viable in the soil for decades, resurface when soil is disturbed.^(1,2,6)

Transmission

Human anthrax infections typically occur through contact with infected livestock, wild animals, or with contaminated animal products such as carcasses, hides, hair, wool, meat or bone meal or items made from these raw materials.^(1,2,6) Transmission can also occur through injection drug use. Although rare, cutaneous anthrax may spread person-to-person through contagious lesions, but other forms of anthrax are not contagious.^(1,2)

The clinical presentation is determined by the route of transmission:

- **Cutaneous anthrax:** Bacteria enter pre-existing lesions, typically via exposed areas.^(1–3)
- **Gastrointestinal anthrax:** Caused by ingestion undercooked contaminated meat or food products. The risk from milk or dairy products is extremely low.^(1,2,8,9)
- **Injection anthrax:** Transmitted through injection of contaminated drugs, particularly heroin.^(1–3)
- **Inhalation anthrax:** Results from inhaling spores, which can be aerosolized during activities like tanning hides or processing wool or bone.^(1–4)

Due to the high stability of anthrax spores, their potential respiratory transmission, and the high mortality of inhalation anthrax, *B. anthracis* is considered a significant bioterrorism agent.^(2,10)

Incubation Period

The incubation period of anthrax varies based on the route of exposure:

- Cutaneous: Typically 5–7 days, but can range from 1–12 days.^(1–3)
- Gastrointestinal: 1–6 days.⁽¹⁾
- Injection: 1–10 days.⁽¹⁾
- Inhalation: due to spore dormancy in lungs, the incubation period may range from 2 days to 6 weeks with potential to extend up to 60 days.^(1–3)

Period of Communicability

Animal products and soil contaminated with anthrax spores can remain infectious for decades.^(1,2) Drainage from untreated human cutaneous anthrax pose a limited risk for direct or indirect (fomite) transmission for several weeks while the lesion is healing.^(1,2)

Host Susceptibility

Susceptibility is universal. The risk of infection is higher for individuals who do not use personal protective equipment (PPE) and:^(1–3,11)

- Have contact with animals or animal products, such as workers processing hides, wool, or bone, veterinarians, and agricultural or wildlife workers, or
- work in a laboratory handling anthrax samples,

The risk is also elevated for individuals who:

- Participate in or attend events where natural hide drums are used, as spores in contaminated products can become aerosolized, or
- Inject, snort, or inhale illicit drugs, particularly heroin.

Incidence

Anthrax occurs worldwide, with higher incidence in agricultural regions, especially in countries without routine animal vaccination programs.⁽⁴⁾ Human outbreaks have been reported in Africa, Asia and Eastern Europe typically linked to handling or consuming infected livestock.⁽¹⁾ *B. anthracis* infections are rare and sporadic in most developed countries, primarily affecting individuals with occupational exposure.⁽¹⁾ Injection anthrax has only been reported among heroin users in Northern Europe.^(1,3)

Anthrax has been notifiable in Canada since 1931.⁽¹²⁾ Disease incidence is available on the [Government of Canada website](#).

Anthrax has been notifiable in Alberta since 1923. The last human case of anthrax in Alberta was reported in 1948. Historical disease incidence is in the [report found here](#).

Since 1967, 20–25 animal outbreaks have been reported with the largest occurring in 2006 in Manitoba and Saskatchewan where 800 animals died on more than 150 premises.⁽¹³⁾ Outbreaks among wild animals, such as bison, have been reported more than once in Wood Buffalo National Park that borders both Alberta and the Northwest Territories.⁽¹³⁾

Resources

For information on animal anthrax in Alberta see [here](#).

Public Health Management

Key Investigation

- Confirm that the individual meets the case definition.
- Obtain history of illness including date of onset of signs and symptoms.
- Determine the possible source of the infection taking into consideration the incubation period, reservoir, and mode of transmission. Assessment should include:
 - occupation, location of work and use of personal protective equipment (PPE), such as:^(1,11)
 - veterinarian or related worker (e.g., necropsy)
 - laboratory workers that handle anthrax specimens,
 - agricultural or farming activities with wildlife or livestock,
 - industrial work involving animal products (e.g., meats, hide and hair)
 - recent contact with deceased animals,
 - play or attend events where natural hide drums are used, and
 - history of injecting, snorting, or inhaling illicit drugs.
- Identify contacts who may have been exposed to the same source (e.g., animal carcass), infected animals or who have had direct contact with cutaneous lesions of a human case.

Management of a Case

- Rapid treatment initiation and availability of advanced, hospitalization is essential for management of disease.^(1–3,7) Consultation with an infectious disease specialist is also recommended.
- Cases pose a limited risk for subsequent person-to-person transmission. They should be isolated in accordance with hospital infection prevention and control guidelines, if applicable.
- Provide information about disease transmission and measures to minimize transmission including the proper disposal of materials from draining lesions (i.e., either incinerated or managed as biohazard waste).

Management of Contacts

- While cases do not pose a substantial risk for subsequent person-to-person transmission, it may be prudent to identify individuals exposed to the same source and advise them to monitor for symptoms for the duration of the [incubation period](#). Should symptoms occur, they should seek urgent advice/care from their health care provider.
- Provide contacts with information about:
 - modes of transmission,
 - signs and symptoms of disease,
 - care of abrasions and wounds,
 - the importance of hand hygiene, and
 - the duration of symptom monitoring (specific timelines based on exposure type – refer to Transmission and Incubation Period)
- [Individuals at risk](#) who have documented or suspected exposure to aerosolized *B. anthracis* should receive antimicrobial post-exposure prophylaxis (PEP) for 60 days.^(1,7,13,14)
 - In combination with antimicrobials, the anthrax vaccine (BioThrax®) may be provided under special circumstances (i.e., confirmed exposure, bioterrorism event, laboratory exposure).⁽¹⁴⁾
 - Consultation with an infectious disease specialist is recommended.
- Ensure that contacts understand reporting requirements for any symptoms and encourage them to report immediately.
- Consider follow-up evaluations to monitor the health status of individuals at risk.
- Provide information about psychological support services for those experiencing anxiety or stress related to their potential exposure.

Preventive Measures

- Provide education to the public about:
 - modes of transmission,
 - signs and symptoms of disease, and
 - care of abrasions and wounds, and importance of hand hygiene, especially after handling animals, animal productions or contact with animal environments.

References

1. Heymann DL, editor. Anthrax. In: Control of Communicable Disease Manual. 21st ed. Washington D.C.: American Public Health Association; 2022. p. 16–24.
2. American Academy of Pediatrics. Anthrax. In: Kimberlin DW, Barnett ED, Lynfield R, Sawyer MH, editors. Red Book: 2021-2024 Report of the Committee on Infectious Diseases. 32nd ed. Itasca, IL: American Academy of Pediatrics; 2021. p. 196–202.
3. Centers for Disease Control and Prevention (CDC). Clinical Care of Anthrax [Internet]. 2024 [cited 2024 Sep 15]. Available from: <https://www.cdc.gov/anthrax/hcp/antibiotics/index.html>
4. Centers for Disease Control and Prevention (CDC). About Anthrax [Internet]. 2024 [cited 2024 Sep 15]. Available from: <https://www.cdc.gov/anthrax/about/index.html>
5. Bush LM., Vazquez-Pertejo MT. Anthrax [Internet]. Merck Manual Professional Version. 2023 [cited 2024 Sep 15]. Available from: https://www.merckmanuals.com/en-ca/professional/infectious-diseases/gram-positive-bacilli/anthrax#Symptoms-and-Signs_v1005892
6. World Health Organization. Anthrax [Internet]. 2016 [cited 2024 Sep 15]. Available from: <https://www.who.int/europe/news-room/questions-and-answers/item/anthrax>
7. Cohen M, Muntner P, Kent CK, Gottardy AJ, Leahy MA, Spriggs SR, et al. CDC Guidelines for the Prevention and Treatment of Anthrax, 2023. Morb Mortal Wkly Rep [Internet]. 2023 [cited 2024 Sep 15];72(6). Available from: <https://www.cdc.gov/mmwr/volumes/72/rr/rr7206a1.htm>
8. Government of Canada. Anthrax – Fact Sheet - Canadian Food Inspection Agency [Internet]. 2013 [cited 2024 Sep 15]. Available from: <https://inspection.canada.ca/animal-health/terrestrial-animals/diseases/reportable/anthrax/fact-sheet/eng/1375205846604/1375206913111>
9. World Health Organization. Annex 6, Contingency plan for the prevention and control of anthrax. In: Anthrax in Humans and Animals [Internet]. 4th ed. 2008. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK310476/#annex6.s3>
10. Centers for Disease Control and Prevention (CDC). Bioterrorism and Anthrax: The Threat [Internet]. 2024 [cited 2024 Sep 15]. Available from: <https://www.cdc.gov/anthrax/bioterrorism/index.html>
11. Centers for Disease Control and Prevention (CDC). Preventing Exposure to Anthrax at Certain Jobs [Internet]. 2024 [cited 2024 Sep 15]. Available from: <https://www.cdc.gov/anthrax/prevention/occupational-exposure.html>
12. Government of Canada. Case definitions: Nationally notifiable diseases [Internet]. 2023 [cited 2024 Sep 15]. Available from: <https://diseases.canada.ca/notifiable/diseases-list>
13. Government of Alberta. Anthrax [Internet]. 2024 [cited 2024 Sep 15]. Available from: <https://www.alberta.ca/anthrax-overview.aspx>
14. Government of Canada. Bacillus anthracis: Infectious substances pathogen safety data sheet [Internet]. 2020 [cited 2024 Sep 15]. Available from: <https://www.canada.ca/en/public-health/services/laboratory-biosafety-biosecurity/pathogen-safety-data-sheets-risk-assessment/bacillus-anthraxis-material-safety-data-sheets-msds.html>